KAIZEN MATHS

Lesson starter

Continuous Random Variables and Density Functions

Standard Models

Name _____

S1

X ~ U(5, 17).Find P(10 < X < 13).

S2

X ~ U(5, 15).Find E(X), Var(X), and the standard deviation.

S3

X has an exponential distribution with rate λ = 0.25.Find P(X > 4).

S4

X ~ Exp(λ) and P(X > 3) = 0.741.Find λ.

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Exit ticket

Continuous Random Variables and Density Functions

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E1

X ~ U(10, 20).Find P(17 < X < 19).

E2

X ~ U(0, 6).Find E(X), Var(X), and the standard deviation.

E3

X has an exponential distribution with rate λ = 0.2.Find P(X < 2).

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Teacher answers & working

Continuous Random Variables and Density Functions · Standard Models

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S1

X ~ U(5, 17).Find P(10 < X < 13).

Answer:

$\frac{1}{4}$

Working:

For a uniform distribution, the density is constant.

Total length = 17 - 5 = 12, so f(x) = $\frac{1}{12}$ on 5 ≤ x ≤ 17.

Set up the probability as an area: P(10 < X < 13) = ∫ from 10 to 13 of $\frac{1}{12}$ dx.

Integrate the constant: ∫ $\frac{1}{12}$ dx = $\frac{x}{12}$.

Substitute the limits: [$\frac{x}{12}$] from 10 to 13 = $\frac{(13-10)}{12}$.

$P = \frac{1}{4} \approx 0.250$

S2

X ~ U(5, 15).Find E(X), Var(X), and the standard deviation.

Answer:

$E(X) = 10, \mathrm{Var}(X) = \frac{25}{3}$

Working:

For X ~ U(a,b), the mean is the midpoint of the interval.

$E(X) = \frac{(a+b)}{2}$.

Substitute a = 5 and b = 15: E(X) = $\frac{(5+15)}{2}$ = 10.

The variance of a uniform distribution depends on the width of the interval.

Use $\mathrm{Var}(X) = \frac{(b-a)^2}{12}$.

$\mathrm{Var}(X) = \frac{(15-5)^2}{12} = \frac{25}{3}$.

SD(X) = sqrt($\frac{25}{3}$) ≈ 2.887.

$E(X) = 10, \mathrm{Var}(X) = \frac{25}{3}$

S3

X has an exponential distribution with rate λ = 0.25.Find P(X > 4).

Answer:

0.368

Working:

For X ~ Exp(λ), integrating the density gives the survival rule P(X > t) = e^(-λt).

Here λ = 0.25 and t = 4.

$P(X > 4) = e^{(-0.25 \times 4)} = 0.368$.

This is already the required probability.

0.368

Teacher answers & working

Continuous Random Variables and Density Functions · Standard Models

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S4

X ~ Exp(λ) and P(X > 3) = 0.741.Find λ.

Answer:

$\lambda \approx 0.100$

Working:

Use the exponential survival function, which comes from integrating the density.

$P(X > t) = e^{(-\lambda t)}$.

$e^{(-3\lambda)} = 0.741$.

Take natural logarithms.

$-3\lambda = \ln(0.741)$.

$\lambda = \frac{-ln(0.741)}{3}$.

$\lambda \approx 0.100$

E1

X ~ U(10, 20).Find P(17 < X < 19).

Answer:

$\frac{1}{5}$

Working:

For a uniform distribution, the density is constant.

Total length = 20 - 10 = 10, so f(x) = $\frac{1}{10}$ on 10 ≤ x ≤ 20.

Set up the probability as an area: P(17 < X < 19) = ∫ from 17 to 19 of $\frac{1}{10}$ dx.

Integrate the constant: ∫ $\frac{1}{10}$ dx = $\frac{x}{10}$.

Substitute the limits: [$\frac{x}{10}$] from 17 to 19 = $\frac{(19-17)}{10}$.

$P = \frac{1}{5} \approx 0.200$

Teacher answers & working

Continuous Random Variables and Density Functions · Standard Models

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E2

X ~ U(0, 6).Find E(X), Var(X), and the standard deviation.

Answer:

$E(X) = 3, Var(X) = 3$

Working:

For X ~ U(a,b), the mean is the midpoint of the interval.

$E(X) = \frac{(a+b)}{2}$.

Substitute a = 0 and b = 6: E(X) = $\frac{(0+6)}{2}$ = 3.

The variance of a uniform distribution depends on the width of the interval.

Use $Var(X) = \frac{(b-a)^2}{12}$.

$Var(X) = \frac{(6-0)^2}{12} = 3$.

SD(X) = sqrt(3) ≈ 1.732.

$E(X) = 3, Var(X) = 3$

E3

X has an exponential distribution with rate λ = 0.2.Find P(X < 2).

Answer:

0.330

Working:

For X ~ Exp(λ), integrating the density gives the survival rule P(X > t) = e^(-λt).

Here λ = 0.2 and t = 2.

$P(X > 2) = e^(-0.2 \times 2) = 0.670$.

$P(X < 2) = 1 - P(X > 2)$.

$P(X < 2) = 1 - 0.670 = 0.330$.

0.330